4.01	**Approximation and estimation**				
4.01a	Rounding	Round numbers to the nearest whole number, ten, hundred, etc. or to a given number of significant figures (sf) or decimal places (dp).	Round answers to an appropriate level of accuracy.		N15
4.01b	Estimation	Estimate or check, without a calculator, the result of a calculation by using suitable approximations. e.g. Estimate, to one significant figure, the cost of 2.8 kg of potatoes at 68p per kg.	Estimate or check, without a calculator, the result of more complex calculations including roots. Use the symbol ≈ appropriately. e.g. $\sqrt{\dfrac{2.9}{0.051 \times 0.62}} \approx 10$		N14

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.
4.01c	Upper and lower bounds		Use inequality notation to write down an error interval for a number or measurement rounded or truncated to a given degree of accuracy. e.g. If $x = 2.1$ rounded to 1 dp, then $2.05 \leqslant x < 2.15$. If $x = 2.1$ truncated to 1 dp, then $2.1 \leqslant x < 2.2$. Apply and interpret limits of accuracy.	Calculate the upper and lower bounds of a calculation using numbers rounded to a known degree of accuracy. e.g. Calculate the area of a rectangle with length and width given to 2 sf. Understand the difference between bounds of discrete and continuous quantities. e.g. If you have 200 cars to the nearest hundred then the number of cars n satisfies: $150 \leqslant n < 250$ and $150 \leqslant n \leqslant 249$.	N15, N16
OCR 5	**Ratio, Proportion and Rates Of Change**				